

Determinant of a Structured Matrix

Topic.

Determinants, row and column operations, structured matrices.

Statement.

Find the determinant of the matrix of order n :

$$A = \begin{pmatrix} 1 & x & x & \cdots & x \\ y & 1 & 0 & \cdots & 0 \\ y & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Solution.

Use elementary determinant properties:

$$R_i \leftarrow R_i + kR_j$$

does not change the determinant.

First remove y from the first column below the first row:

$$\forall i = 2, \dots, n : \quad R_i \leftarrow R_i - yR_1.$$

This operation does not change the determinant.

Then

$$\det A = \det A',$$

where

$$A' = \begin{pmatrix} 1 & x & x & \cdots & x \\ 0 & 1 - yx & -yx & \cdots & -yx \\ 0 & -yx & 1 - yx & \cdots & -yx \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & -yx & -yx & \cdots & 1 - yx \end{pmatrix}.$$

Expanding by the first column, we get

$$\det A = \det B,$$

where B is the right lower block of size $(n-1) \times (n-1)$:

$$B = \begin{pmatrix} 1 - yx & -yx & \cdots & -yx \\ -yx & 1 - yx & \cdots & -yx \\ \vdots & \vdots & \ddots & \vdots \\ -yx & -yx & \cdots & 1 - yx \end{pmatrix}.$$

Let

$$m = n - 1.$$

Then B is an $m \times m$ matrix.

Now perform row operations:

$$\forall i = 2, \dots, m : \quad R_i \leftarrow R_i - R_1.$$

The determinant does not change.

We get

$$B' = \begin{pmatrix} 1 - yx & -yx & -yx & \cdots & -yx \\ -1 & 1 & 0 & \cdots & 0 \\ -1 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Now make a column operation:

$$C_1 \leftarrow C_1 + C_2 + \cdots + C_m.$$

The determinant does not change.

In the first row, the first element becomes

$$(1 - yx) + (m - 1)(-yx) = 1 - yx - (m - 1)yx = 1 - mxy.$$

In rows $2, \dots, m$, the first column becomes zero because there is -1 in the first column and exactly one 1 among columns $2, \dots, m$.

Thus the matrix becomes upper triangular:

$$B'' = \begin{pmatrix} 1 - mxy & -yx & -yx & \cdots & -yx \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Therefore,

$$\det B = \det B'' = (1 - mxy) \cdot 1 \cdot \dots \cdot 1.$$

Hence

$$\det B = 1 - mxy.$$

Since

$$m = n - 1,$$

we obtain

$$\boxed{\det A = 1 - (n - 1)xy.}$$